

HBM2E Single Memory Core Die — Channel & Bank Layout

Each core die hosts 2 channels; each channel has 2 pseudo channels (PCs) of 16 banks each

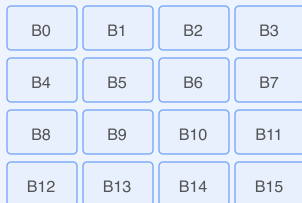
Memory Core Die (one layer in the HBM2E stack)

Channel A

128-bit data bus (DQ[127:0]), own CK, R, C

Pseudo Channel 0

DQ[63:0] — 64 bits
16 Banks (B0–B15)



Row buffer (sense amps)

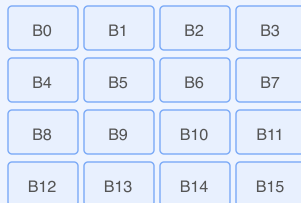
Column decoder

Shared: CK (clock), R (row command), C (column command)

Per-channel I/O: RDQS, WDQS, DBI, DM, PAR (204 signals total)

Pseudo Channel 1

DQ[127:64] — 64 bits
16 Banks (B0–B15)



Row buffer (sense amps)

Column decoder

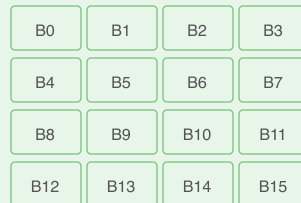
↓ ~5,000 TSVs per die connect to base die (I/F die) below ↓

Channel C

128-bit data bus (DQ[127:0]), own CK, R, C

Pseudo Channel 0

DQ[63:0] — 64 bits
16 Banks (B0–B15)



Row buffer (sense amps)

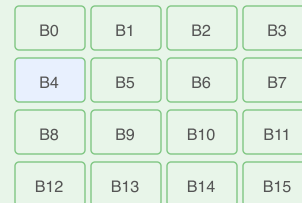
Column decoder

Shared: CK (clock), R (row command), C (column command)

Per-channel I/O: RDQS, WDQS, DBI, DM, PAR (204 signals total)

Pseudo Channel 1

DQ[127:64] — 64 bits
16 Banks (B0–B15)



Row buffer (sense amps)

Column decoder

Each bank contains DRAM cell arrays organized in rows x columns.

A row activation opens one row into the sense amplifiers (row buffer); column access selects from the open row.

Both PCs within a channel share clock and command inputs but have independent 64-bit data paths → effectively 2x parallelism per channel.

Channel pairs are assigned per die: e.g. Die 0 hosts Channels A+C, Die 1 hosts Channels B+D (see stack diagram)